

Information Security Laboratory Setup Guide

This is suitable for UG Information Security / Cyber Security Labs.

INFORMATION SECURITY LAB

Complete Lab Environment Setup Guide

PART 1 – Install Oracle VirtualBox

VirtualBox allows you to run Kali Linux as a virtual machine.

Step-1: Download

Open browser → Go to
<https://www.virtualbox.org>

Download:

- **VirtualBox platform package**
 - **Extension Pack**
-

Step-2: Install VirtualBox

On Windows

1. Double click VirtualBox-x.x.x.exe
2. Click **Next** → **Next** → **Install**
3. Allow network drivers
4. Click **Finish**

On Ubuntu

```
sudo apt update
```

```
sudo apt install virtualbox
```

Step-3: Install Extension Pack

Open VirtualBox →
File → Tools → Extensions → Install
Select downloaded .vbox-extpack → Install

PART 2 – Install Kali Linux in VirtualBox

Kali Linux is required for **VAPT, SIEM testing, malware analysis**.

Step-1: Download Kali VirtualBox Image

<https://www.kali.org/get-kali/#kali-virtual-machines>

Download **Kali Linux VirtualBox Image**

Step-2: Import into VirtualBox

1. Open VirtualBox
 2. Click **File** → **Import Appliance**
 3. Select downloaded .ova file
 4. Click **Import**
-

Step-3: Start Kali Linux

Login:

Username: kali

Password: kali

Step-4: Update Kali

sudo apt update

sudo apt upgrade

PART 3 – Install Java (JDK 17)

Java is used for **AES, RSA, SSL, Diffie-Hellman, Hashing**.

On Kali / Ubuntu

sudo apt install openjdk-17-jdk

java --version

PART 4 – Install MySQL Database

Used for:

- User records
-

- Files
- Exam data
- Transactions

```
sudo apt install mysql-server
```

```
sudo mysql_secure_installation
```

Login:

```
mysql -u root -p
```

Create lab database:

```
CREATE DATABASE infosec;
```

```
CREATE USER 'labuser'@'localhost' IDENTIFIED BY 'lab123';
```

```
GRANT ALL PRIVILEGES ON infosec.* TO 'labuser'@'localhost';
```

PART 5 – Install Visual Studio Code

Used to write Java Client and Server programs.

On Kali / Ubuntu

```
sudo apt install wget gpg
```

```
wget -qO- https://packages.microsoft.com/keys/microsoft.asc | gpg --dearmor > packages.gpg
```

```
sudo install -o root -g root -m 644 packages.gpg /usr/share/keyrings/
```

```
sudo sh -c 'echo "deb [arch=amd64 signed-by=/usr/share/keyrings/packages.gpg]
https://packages.microsoft.com/repos/code stable main" > /etc/apt/sources.list.d/vscode.list'
```

```
sudo apt update
```

```
sudo apt install code
```

Run:

```
code
```

Install extensions:

- Java Extension Pack
- MySQL
- GitHub

PART 6 – Essential Security Tools

```
sudo apt install wireshark nmap nikto ufw clamav openssl ldap-utils
```

PART 7 – Enable Firewall

```
sudo ufw enable
```

```
sudo ufw allow 8443
```

```
sudo ufw deny 80
```

PART 8 – Enable Wireshark

```
sudo dpkg-reconfigure wireshark-common
```

```
sudo usermod -aG wireshark kali
```

Logout and login again.

PART 9 – Verify Tools

Tool	Command
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Java	java -version
------	---------------

MySQL	mysql -u labuser -p
-------	---------------------

Kali tools	nmap --version
------------	----------------

Wireshark	wireshark
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Firewall	sudo ufw status
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What Students Now Have

They can perform:

- ✓ AES & RSA coding
 - ✓ SSL communication
 - ✓ Packet capture
 - ✓ LDAP authentication
 - ✓ Firewall testing
 - ✓ Malware detection
 - ✓ Vulnerability scanning
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 **This environment fully supports:**

Your **UCSPC0633 – Information Security Laboratory**
and all **12 experiments + capstone projects**.

Batch 1: Secure Online Examination System

Batch 2: Banking Transaction Security System

Sample Java Client and Server Program

This Java client-server program uses TCP sockets to send an array of integers from client to server, where the server sorts it using `Arrays.sort()` and sends it back for display. Run the server first, then the client. The client prompts for array size and elements (not fixed at 5).

Server Code

```
java

import java.io.*;
import java.net.*;
import java.util.Arrays;

class Server {

    public static void main(String args[]) throws Exception {

        ServerSocket ss = new ServerSocket(7777);

        Socket s = ss.accept();

        System.out.println("connected...");

        DataInputStream din = new DataInputStream(s.getInputStream());
        DataOutputStream dout = new DataOutputStream(s.getOutputStream());

        int n = din.readInt();

        int a[] = new int[n];

        System.out.println("Receiving Data...");

        for(int i = 0; i < n; i++) {

            a[i] = din.readInt();

        }

        System.out.println("Data Received. Original: " + Arrays.toString(a));

        System.out.println("Sorting Data...");

        Arrays.sort(a);

        for(int i = 0; i < n; i++) {

            dout.writeInt(a[i]);

        }

        System.out.println("Data Sent Successfully: " + Arrays.toString(a));

        s.close();

    }

}
```

```

        ss.close();
    }
}

```

The server listens on port 7777, reads the array size n, receives n integers, sorts them ascending, and sends back the sorted array.

Client Code

java

```
import java.io.*;
```

```
import java.net.*;
```

```
import java.util.Scanner;
```

```
class Client {
```

```
    public static void main(String args[]) throws Exception {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter size of array: ");
```

```
        int n = sc.nextInt();
```

```
        int a[] = new int[n];
```

```
        System.out.println("Enter " + n + " numbers:");
```

```
        for(int i = 0; i < n; i++) {
```

```
            a[i] = sc.nextInt();
```

```
        }
```

```
        Socket s = new Socket("localhost", 7777);
```

```
        DataInputStream din = new DataInputStream(s.getInputStream());
```

```
        DataOutputStream dout = new DataOutputStream(s.getOutputStream());
```

```
        dout.writeInt(n);
```

```
        for(int i = 0; i < n; i++) {
```

```
            dout.writeInt(a[i]);
```

```
        }
```

```
        System.out.println("Original array: ");
```

```
        for(int i = 0; i < n; i++) {
```

```
            System.out.print(a[i] + " ");
```

```
}  
System.out.println("\nSorted array received from server:");  
for(int i = 0; i < n; i++) {  
    a[i] = din.readInt();  
    System.out.print(a[i] + " ");  
}  
System.out.println();  
s.close();  
sc.close();  
}  
}
```